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Provadis School of International Management and Technology

Exposé

Examining the Feasibility of Machine Learning-Based LogP Prediction on an ESP32 Microcontroller

A Proof-of-Concept Implementation and Performance Analysis

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Contents

Glossary

ESP32 Espressif Systems' low-cost, low-power system on a chip (SoC) with integrated Wi-Fi and Bluetooth capabilities. 1–3

Fingerprint A binary vector representation of a molecule, where each bit represents the presence or absence of a particular substructure or feature in the molecule. 1, 2

IoT Internet of Things. 1

logP Logarithm of the octanol-water partition coefficient. A measure of a compound's hydrophobicity, or how well it dissolves in fats versus water. 1, 2

QR Code Quick Response Code, a type of matrix barcode (or two-dimensional barcode) that can store information such as a URL, text, or other data. 1

RF Random Forest, an ensemble learning method for classification and regression tasks that constructs multiple decision trees during training. 2

SMILES Simplified Molecular Input Line Entry System, a notation that allows a user to represent a chemical structure as a string of ASCII characters. 2

SVM Support Vector Machine, a supervised machine learning algorithm used for classification and regression tasks that finds the hyperplane that best separates the classes in the feature space. 2

TFLite TensorFlow Lite, a framework for deploying machine learning models on mobile and embedded devices. 2

ZINC20 ZINC Is Not Commercial (2020), a free database of commercially available compounds for virtual screening. 2

Exposé

Problem Statement

LogP, or the logarithm of the octanol-water partition coefficient, is a key physicochemical property that describes the lipophilicity (ability to dissolve in fats, oils, and other non-polar solvents) of a compound. It is defined as the ratio of the concentration of a compound in a mixture of two immiscible solvents (usually octanol and water) at equilibrium. The more lipophilic a compound is, the higher its logP value, which can influence its absorption, distribution, metabolism, and excretion (ADME) properties in biological systems.

The accurate prediction of logP is crucial in various fields, including drug discovery and environmental chemistry. Traditional computational methods for logP prediction can be resource-intensive because the rely on storing large lists of fragments and their contributions to the overall logP value. This can make it challenging to deploy such models on resource-constrained devices like microcontrollers. Other, even more accurate methods, such as those based on quantum chemistry calculations, are even more computationally expensive and thus not suitable for deployment on microcontrollers. This project aims to explore the feasibility of implementing machine learning-based logP prediction (based on a molecule's Fingerprint) on an ESP32 microcontroller, which is widely used in IoT applications.

The reason for exploring the ability to predict logP on a microcontroller is to enable real-time, on-site predictions in various applications, such as environmental monitoring or portable drug discovery tools. By offloading the prediction task away from a host computer and onto a microcontroller, one can potentially reduce latency in systems that require immediate response, such as a microfluidic dispensing device that needs to make decisions based on the logP of a compound in real-time.

Since the molecular Fingerprint is a bit vector that could be encoded as a QR Code, the microcontroller could also be used in scenarios where a user can scan said code printed on the sample in a laboratory or field setting with a portable device and receive an immediate prediction of the logP value, and if this approach seems feasible and accurate enough, it could be extended to predicting other properties as well, without needing to access a cloud-based service or a powerful computer.

Also, by decoupling the Fingerprint generation (a deterministic process on canonical SMILES strings) and the prediction (a non-deterministic process owing to the nature of machine learning models), and by binding the prediction to a specific piece of hardware (a "black box", as it were), one can make the entire system more modular and easier to debug. As an added bonus, the host computer wouldn't need to have any machine learning libraries installed.

Objectives

The primary objectives of this project are:

- To develop a machine learning model for logP prediction using a suitable dataset.
- To optimize the model for deployment on an ESP32 microcontroller, considering constraints such as memory and computational power.
- To evaluate the performance of the implemented model in terms of accuracy and efficiency.

Methodology

The project will follow these steps:

1. Data Collection: Gather a dataset of chemical compounds with known logP values. The ZINC20 database will be used as a primary source for this data.
2. Model Development: Train a machine learning model (e.g., RF, SVM, Neural Network) using the dataset and the TFLite framework for microcontroller deployment.
3. Model Optimization: Use techniques such as quantization and pruning to reduce the model size for deployment on the ESP32.
4. Implementation: Deploy the optimized model on the ESP32 microcontroller and develop an interface for inputting chemical compound data.
5. Evaluation: Assess the model's performance using metrics such as mean absolute error (MAE) and inference time on the ESP32.

Planned Structure

The paper will likely be structured as follows:

- Introduction
- Research Question and Objectives
- Literature Review
- Methodology
- Results and Discussion
- Limitations
- Conclusion and Future Work
- References
- AI Declaration
- Declaration of Authorship

Projected Timeline

- 2026-03-02: Exposé submission
- 2026-03-10: Literature review completion
- 2026-03-15: Data collection and preprocessing
- 2026-03-17: Model development, training, and deployment on ESP32
- 2026-03-31: Submission of the final paper

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